

Администрирование локальных сетей

Конфигурирование VLAN

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Цели и задачи работы

Получить основные навыки по настройке VLAN на коммутаторах сети.

Топология сети

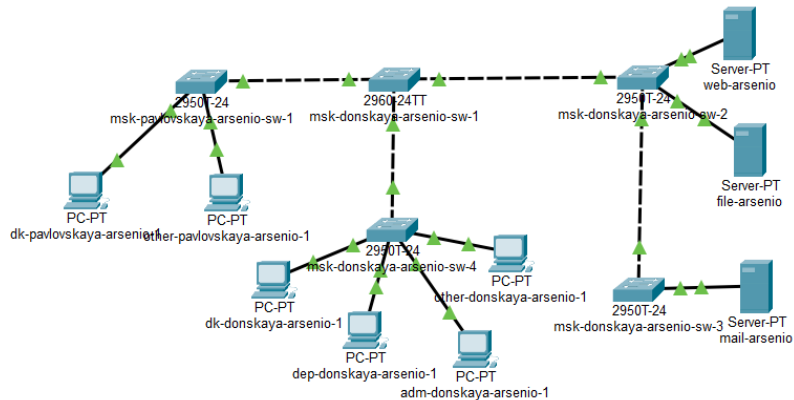


Рис. 1: Построенная топология сети в Packet Tracer

Настройка VTP и VLAN

VTP Server и создание VLAN

```
msk-donskaya-arsenio-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-arsenio-sw-1(config)#vtp mode server
Device mode already VTP SERVER.
msk-donskaya-arsenio-sw-1(config)#vtp dom donsкаya
Changing VTP domain name from NULL to donsкаya
msk-donskaya-arsenio-sw-1(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to
donsкаya.

msk-donskaya-arsenio-sw-1(config)#vtp pass cisco
Setting device VLAN database password to cisco
msk-donskaya-arsenio-sw-1(config)#vlan 2
msk-donskaya-arsenio-sw-1(config-vlan)#name management
msk-donskaya-arsenio-sw-1(config-vlan)#vlan 3
msk-donskaya-arsenio-sw-1(config-vlan)#name servers
msk-donskaya-arsenio-sw-1(config-vlan)#vlan 101
msk-donskaya-arsenio-sw-1(config-vlan)#name dk
msk-donskaya-arsenio-sw-1(config-vlan)#vlan 102
msk-donskaya-arsenio-sw-1(config-vlan)#name departments
msk-donskaya-arsenio-sw-1(config-vlan)#vlan 103
msk-donskaya-arsenio-sw-1(config-vlan)#name adm
msk-donskaya-arsenio-sw-1(config-vlan)#vlan 104
msk-donskaya-arsenio-sw-1(config-vlan)#name other
msk-donskaya-arsenio-sw-1(config-vlan)#int f0/24
msk-donskaya-arsenio-sw-1(config-if)#sw mode trunk
msk-donskaya-arsenio-sw-1(config-if)#int f0/1
msk-donskaya-arsenio-sw-1(config-if)#sw mode trunk
msk-donskaya-arsenio-sw-1(config-if)#int r g0/1 - g0/2
msk-donskaya-arsenio-sw-1(config-if-range)#sw mode trunk

msk-donskaya-arsenio-sw-1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

msk-donskaya-arsenio-sw-1(config-if-range)#^Z
msk-donskaya-arsenio-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-sw-1#wr mem
Building configuration...
[OK]
```

```
Password:
msk-donskaya-arsenio-sw-2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-arsenio-sw-2(config)#vtp mode cli
Setting device to VTP CLIENT mode.
msk-donskaya-arsenio-sw-2(config)#vtp dom donskeya
Changing VTP domain name from NULL to donskeya
msk-donskaya-arsenio-sw-2(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to
donskeya.

msk-donskaya-arsenio-sw-2(config)#vtp pass cisco
Setting device VLAN database password to cisco
msk-donskaya-arsenio-sw-2(config)#int r f0/1 - f0/2
msk-donskaya-arsenio-sw-2(config-if-range)#sw mode trunk

msk-donskaya-arsenio-sw-2(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

msk-donskaya-arsenio-sw-2(config-if-range)#no sw mode trunk
msk-donskaya-arsenio-sw-2(config-if-range)#sw mode acc
msk-donskaya-arsenio-sw-2(config-if-range)#sw acc vlan 3
msk-donskaya-arsenio-sw-2(config-if-range)#
msk-donskaya-arsenio-sw-2(config-if-range)#int r g0/1 - g0/2
msk-donskaya-arsenio-sw-2(config-if-range)#no sw mode trunk
msk-donskaya-arsenio-sw-2(config-if-range)#
```

VTP Client на msk-donskaya-sw-3

```
msk-donskaya-arsenio-sw-3>en
Password:
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

msk-donskaya-arsenio-sw-3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-arsenio-sw-3(config)#vtp mode cli
Setting device to VTP CLIENT mode.
msk-donskaya-arsenio-sw-3(config)#vtp dom donsкаaya
Domain name already set to donsкаaya.
msk-donskaya-arsenio-sw-3(config)#vtp pass cisco
Setting device VLAN database password to cisco
msk-donskaya-arsenio-sw-3(config)#int g0/1
msk-donskaya-arsenio-sw-3(config-if)#sw mode trunk
msk-donskaya-arsenio-sw-3(config-if)#int r f0/1 - f0/2
msk-donskaya-arsenio-sw-3(config-if-range)#sw mode acc
msk-donskaya-arsenio-sw-3(config-if-range)#sw acc vlan 3
msk-donskaya-arsenio-sw-3(config-if-range)#^Z
msk-donskaya-arsenio-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-sw-3#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-sw-3#
```

Настройка портов доступа на SW-4

```
msk-donskaya-arsenio-sw-4#
msk-donskaya-arsenio-sw-4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-arsenio-sw-4(config)#
msk-donskaya-arsenio-sw-4(config)#vtp mode cli
Setting device to VTP CLIENT mode.
msk-donskaya-arsenio-sw-4(config)#vtp dom donskaya
Domain name already set to donskaya.
msk-donskaya-arsenio-sw-4(config)#vtp pass cisco
Setting device VLAN database password to cisco
msk-donskaya-arsenio-sw-4(config)#
msk-donskaya-arsenio-sw-4(config)#int g0/1
msk-donskaya-arsenio-sw-4(config-if)#sw mode trunk
msk-donskaya-arsenio-sw-4(config-if)#int r f0/1 - f0/5
msk-donskaya-arsenio-sw-4(config-if-range)#sw mode acc
msk-donskaya-arsenio-sw-4(config-if-range)#sw acc vlan 101
msk-donskaya-arsenio-sw-4(config-if-range)#int r f0/6 - f0/10
msk-donskaya-arsenio-sw-4(config-if-range)#sw mode acc
msk-donskaya-arsenio-sw-4(config-if-range)#sw acc vlan 102
msk-donskaya-arsenio-sw-4(config-if-range)#int r f0/11 - f0/15
msk-donskaya-arsenio-sw-4(config-if-range)#sw mode acc
msk-donskaya-arsenio-sw-4(config-if-range)#sw acc vlan 103
msk-donskaya-arsenio-sw-4(config-if-range)#int r f0/16 - f0/20
msk-donskaya-arsenio-sw-4(config-if-range)#sw mode acc
msk-donskaya-arsenio-sw-4(config-if-range)#sw acc vlan 104
msk-donskaya-arsenio-sw-4(config-if-range)#^Z
msk-donskaya-arsenio-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-sw-4#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-sw-4#
```

Настройка портов доступа на pavlovskaya-sw-1

```
msk-pavlovskaya-arsenio-sw-1>en
Password:
msk-pavlovskaya-arsenio-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-pavlovskaya-arsenio-sw-1(config)#vtp mode cli
Setting device to VTP CLIENT mode.
msk-pavlovskaya-arsenio-sw-1(config)#vtp dom donskaya
Domain name already set to donskaya.
msk-pavlovskaya-arsenio-sw-1(config)#vtp pass cisco
Setting device VLAN database password to cisco
msk-pavlovskaya-arsenio-sw-1(config)#int f0/24
msk-pavlovskaya-arsenio-sw-1(config-if)#sw mode trunk
msk-pavlovskaya-arsenio-sw-1(config-if)#int r f0/1 - f0/15
msk-pavlovskaya-arsenio-sw-1(config-if-range)#sw mode acc
msk-pavlovskaya-arsenio-sw-1(config-if-range)#sw acc vlan 101
msk-pavlovskaya-arsenio-sw-1(config-if-range)#int r f0/20 - f0/22
msk-pavlovskaya-arsenio-sw-1(config-if-range)#sw mode acc
msk-pavlovskaya-arsenio-sw-1(config-if-range)#sw acc vlan 104
msk-pavlovskaya-arsenio-sw-1(config-if-range)#^Z
msk-pavlovskaya-arsenio-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-pavlovskaya-arsenio-sw-1#wr mem
Building configuration...
[OK]
msk-pavlovskaya-arsenio-sw-1#
```

Статическая адресация

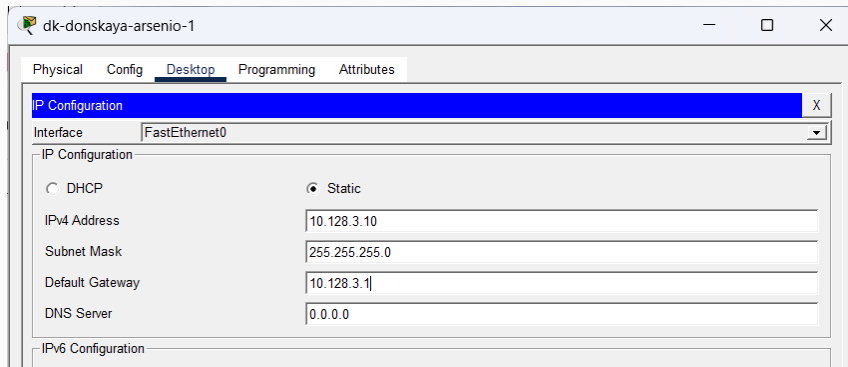


Рис. 7: Настройка IP-адреса на dk-donskaya-arsenio-1

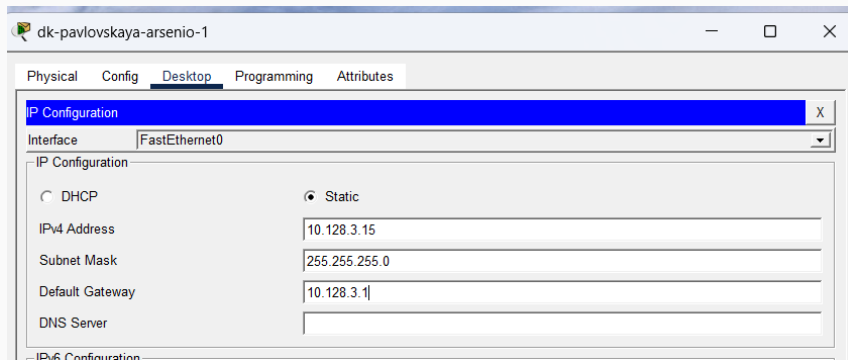


Рис. 8: Настройка IP-адреса на dk-pavlovskaya-arsenio-1

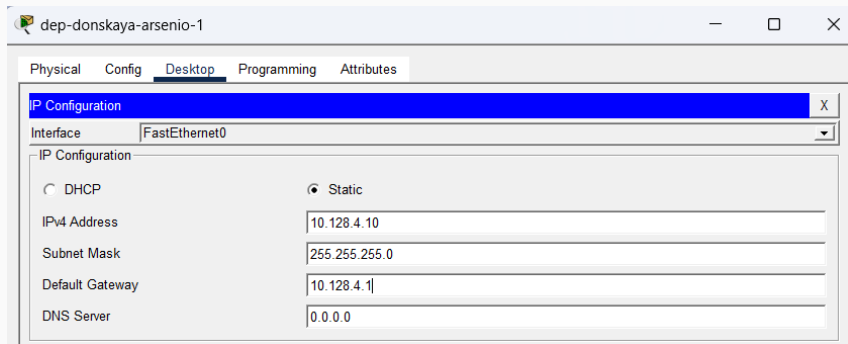


Рис. 9: Настройка IP-адреса на dep-donskaya-arsenio-1

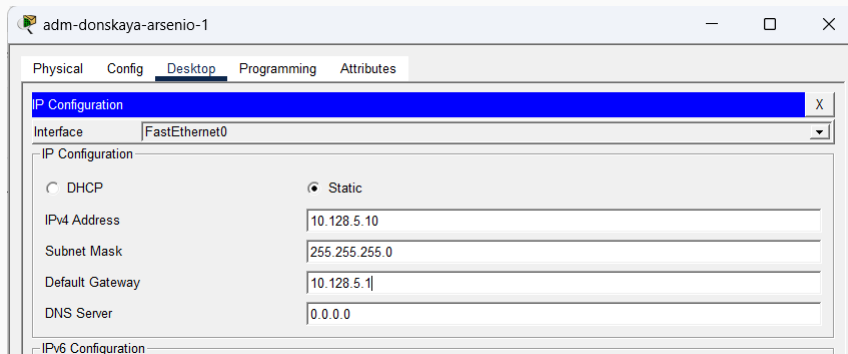


Рис. 10: Настройка IP-адреса на adm-donskaya-arsenio-1

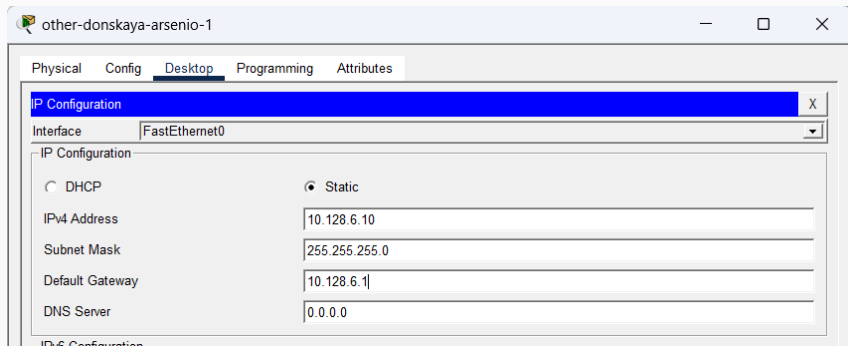


Рис. 11: Настройка IP-адреса на other-donskaya-arsenio-1

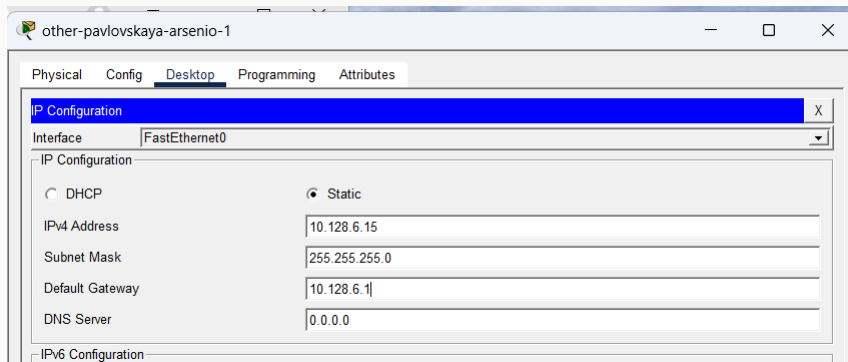


Рис. 12: Настройка IP-адреса на other-pavlovskaya-arsenio-1

Адресация серверов

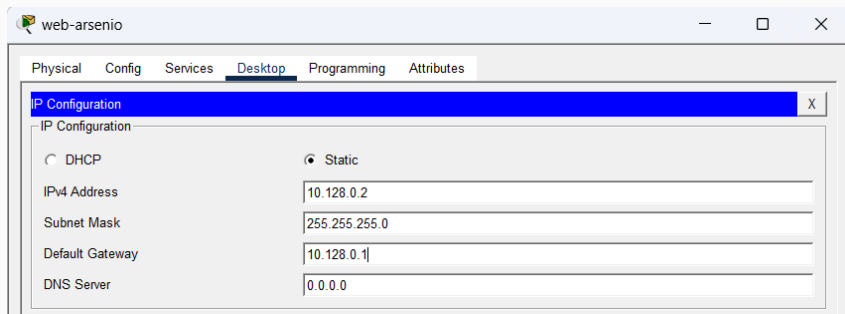


Рис. 13: Настройка IP-адреса на сервере web-arsenio

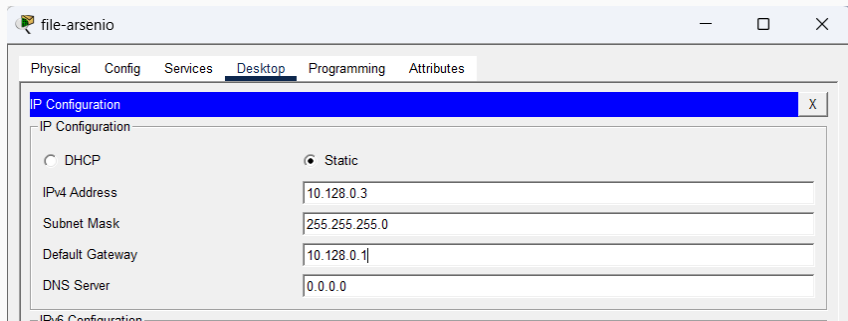


Рис. 14: Настройка IP-адреса на сервере file-arsenio

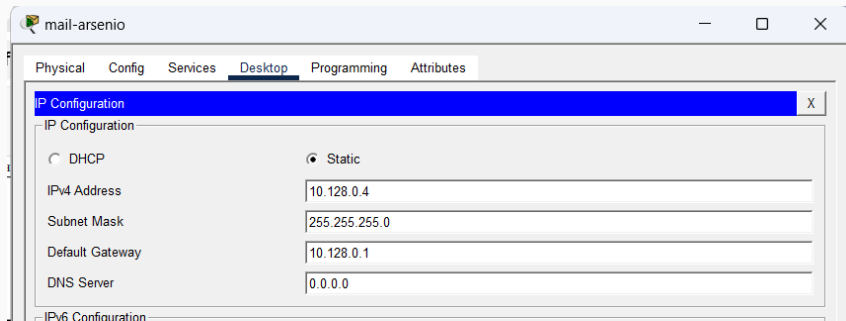
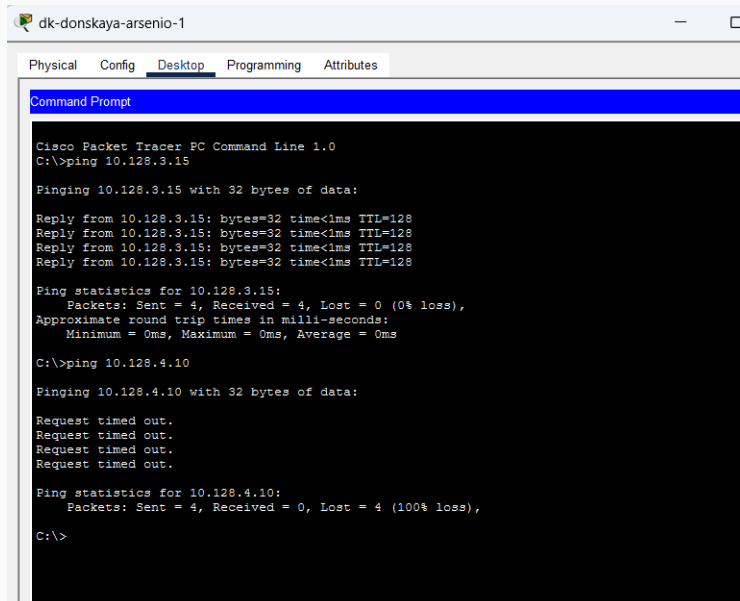


Рис. 15: Настройка IP-адреса на сервере mail-arsenio

Проверка связности

Проверка VLAN 101



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'dk-donskaya-arsenio-1'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The 'Command Prompt' tab is active, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.3.15

Pinging 10.128.3.15 with 32 bytes of data:

Reply from 10.128.3.15: bytes=32 time<1ms TTL=128
Reply from 10.128.3.15: bytes=32 time<1ms TTL=128
Reply from 10.128.3.15: bytes=32 time<1ms TTL=128
Reply from 10.128.3.15: bytes=32 time<1ms TTL=128

Ping statistics for 10.128.3.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.4.10

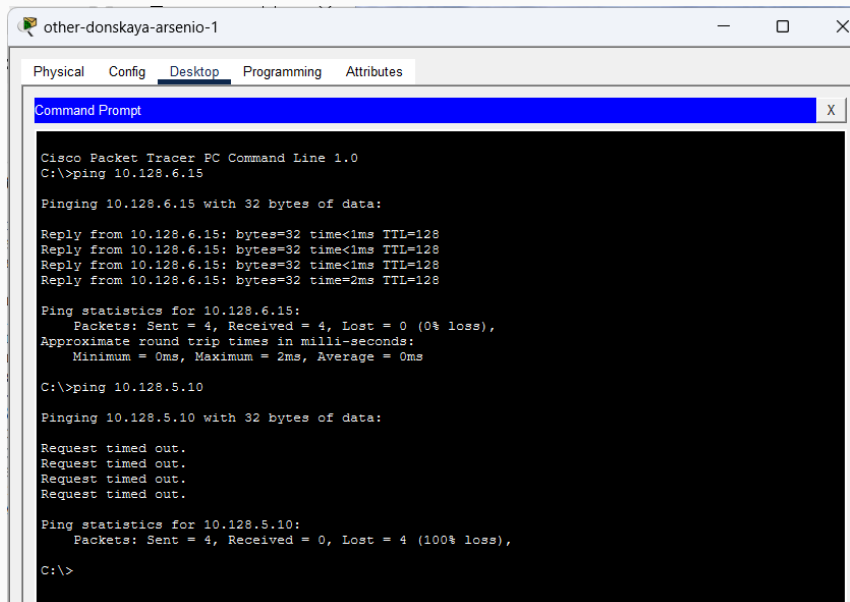
Pinging 10.128.4.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.128.4.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Проверка VLAN 104



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'other-donskaya-arsenio-1'. The 'Desktop' tab is selected. The command prompt shows a successful ping to 10.128.6.15 and a failed ping to 10.128.5.10.

```
other-donskaya-arsenio-1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.6.15

Pinging 10.128.6.15 with 32 bytes of data:

Reply from 10.128.6.15: bytes=32 time<1ms TTL=128
Reply from 10.128.6.15: bytes=32 time<1ms TTL=128
Reply from 10.128.6.15: bytes=32 time<1ms TTL=128
Reply from 10.128.6.15: bytes=32 time=2ms TTL=128

Ping statistics for 10.128.6.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 10.128.5.10

Pinging 10.128.5.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.128.5.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Проверка таблиц VLAN

show vlan на VTP Server

```
msk-donskaya-arsenio-sw-1#sh vlan
```

VLAN Name	Status	Ports
1 default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
2 management	active	
3 servers	active	
101 dk	active	
102 departaments	active	
103 adm	active	
104 other	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
2	enet	100002	1500	-	-	-	-	-	0	0
3	enet	100003	1500	-	-	-	-	-	0	0
101	enet	100101	1500	-	-	-	-	-	0	0
102	enet	100102	1500	-	-	-	-	-	0	0
103	enet	100103	1500	-	-	-	-	-	0	0
104	enet	100104	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
------	------	------	-----	--------	--------	----------	-----	----------	--------	--------

Remote SPAN VLANs

Primary	Secondary	Type	Ports
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```
msk-donskaya-arsenio-sw-1#
```

```
msk-donskaya-arsenio-sw-1#
```

show vlan на sw-2

```
msk-donskaya-arsenio-sw-2#sh vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24
2	management	active	
3	servers	active	Fa0/1, Fa0/2
101	dk	active	
102	departaments	active	
103	adm	active	
104	other	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
2	enet	100002	1500	-	-	-	-	-	0	0
3	enet	100003	1500	-	-	-	-	-	0	0
101	enet	100101	1500	-	-	-	-	-	0	0
102	enet	100102	1500	-	-	-	-	-	0	0
103	enet	100103	1500	-	-	-	-	-	0	0
104	enet	100104	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
------	------	------	-----	--------	--------	----------	-----	----------	--------	--------

Remote SPAN VLANs

Primary	Secondary	Type	Ports
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show vlan на sw-3

```
msk-donskaya-arsenio-sw-3#sh vlan
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
2	management	active	
3	servers	active	Fa0/1, Fa0/2
101	dk	active	
102	departaments	active	
103	adm	active	
104	other	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
2	enet	100002	1500	-	-	-	-	-	0	0
3	enet	100003	1500	-	-	-	-	-	0	0
101	enet	100101	1500	-	-	-	-	-	0	0
102	enet	100102	1500	-	-	-	-	-	0	0
103	enet	100103	1500	-	-	-	-	-	0	0
104	enet	100104	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
Remote SPAN VLANs										
Primary	Secondary	Type	Ports							

show vlan на sw-4

```
msk-donskaya-arsenio-sw-4#sh vlan
```

VLAN Name	Status	Ports
1 default	active	Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/2
2 management	active	
3 servers	active	
101 dk	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5
102 departaments	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10
103 adm	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15
104 other	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
2	enet	100002	1500	-	-	-	-	-	0	0
3	enet	100003	1500	-	-	-	-	-	0	0
101	enet	100101	1500	-	-	-	-	-	0	0
102	enet	100102	1500	-	-	-	-	-	0	0
103	enet	100103	1500	-	-	-	-	-	0	0
104	enet	100104	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
------	------	------	-----	--------	--------	----------	-----	----------	--------	--------

Remote SPAN VLANs

Primary	Secondary	Type	Ports
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show vlan на pavlovskaya-sw-1

```
msk-pavlovskaya-arsenio-sw-1#sh vlan
```

VLAN Name	Status	Ports
1 default	active	Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/23, Gig0/1, Gig0/2
2 management	active	
3 servers	active	
101 dk	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15
102 departaments	active	
103 adm	active	
104 other	active	Fa0/20, Fa0/21, Fa0/22
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
2	enet	100002	1500	-	-	-	-	-	0	0
3	enet	100003	1500	-	-	-	-	-	0	0
101	enet	100101	1500	-	-	-	-	-	0	0
102	enet	100102	1500	-	-	-	-	-	0	0
103	enet	100103	1500	-	-	-	-	-	0	0
104	enet	100104	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
------	------	------	-----	--------	--------	----------	-----	----------	--------	--------

```
Remote SPAN VLANs
```

Primary	Secondary	Type	Ports
---------	-----------	------	-------

```
msk-pavlovskaya-arsenio-sw-1#
```

Анализ ICMP в режиме Simulation

Передача ICMP-пакета

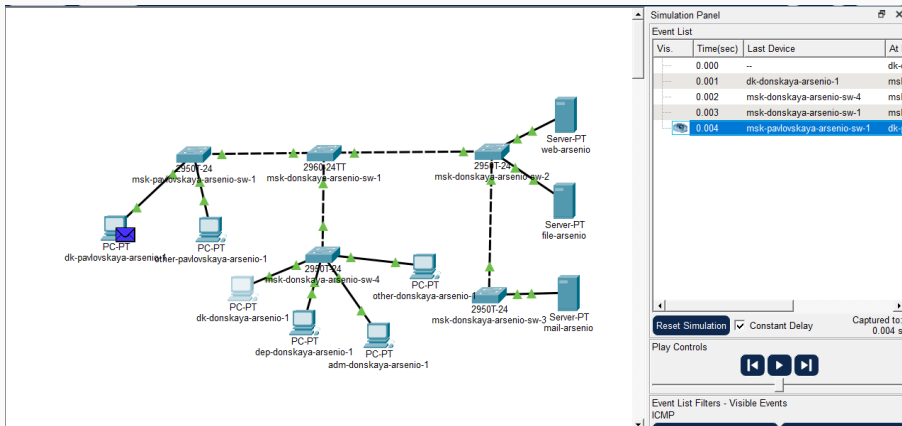
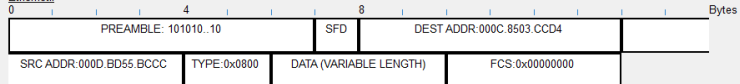


Рис. 23: Передача ICMP-пакета в режиме Simulation

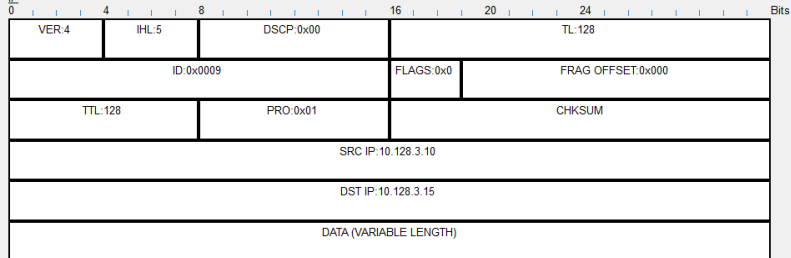
ICMP Echo Request

PDU Formats

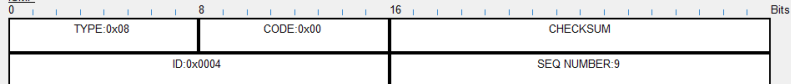
Ethernet II



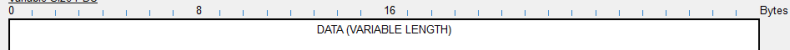
IP



ICMP



Variable Size PDU



Обратная передача ICMP-пакета

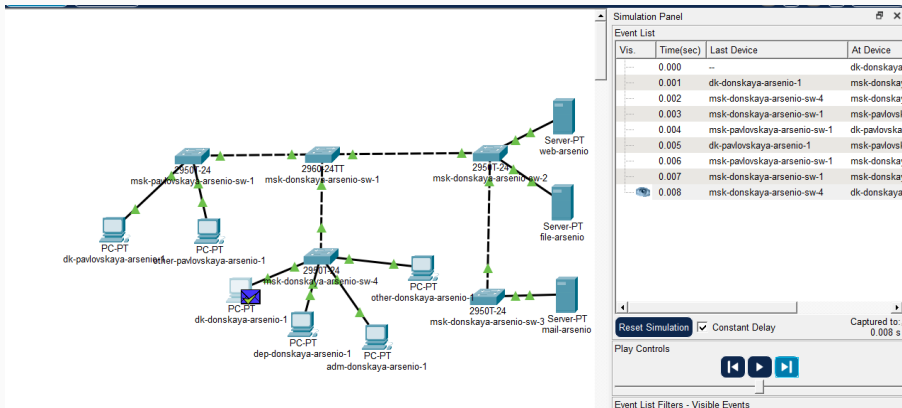
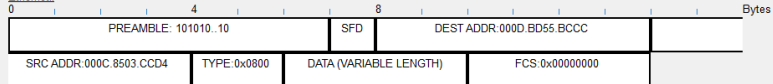


Рис. 25: Обратная передача ICMP-пакета в режиме Simulation

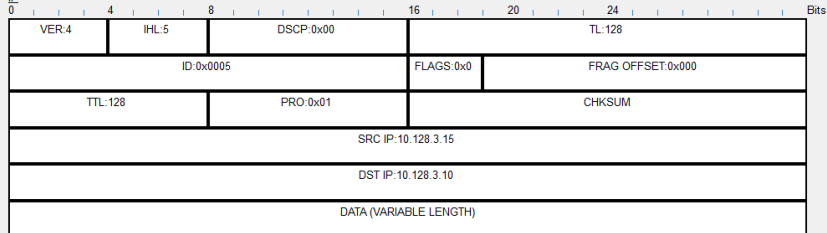
ICMP Echo Reply

PDU Formats

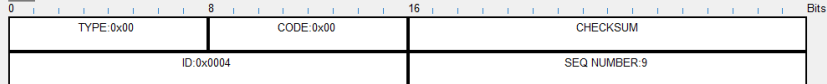
EthernetII



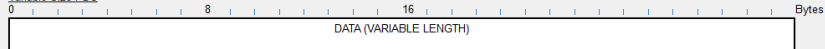
IP



ICMP



Variable Size PDU



Выводы

В ходе лабораторной работы была построена коммутируемая локальная сеть с несколькими VLAN.

Были настроены VTP-домен, Trunk-порты, порты доступа, статические IP-адреса и проверена связность устройств внутри VLAN.

Устройства из одной VLAN успешно обменивались ICMP-пакетами, а устройства из разных VLAN оказались изолированы друг от друга без настройки межсетевой маршрутизации.